Band-1 Computational Experiment: Minimal Superpermutation Lengths (small n)

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim a new record for $n \geq 6$. Exact minimal lengths $s(n)$ are **box-certified** for $n = 1..4$ by exhaustive BFS, and the classical $n = 5$ length-153 witness is certificate-checked (minimality by Chaffin 2014 literature). The next open exact value is $n = 6$ ($868 \leq s(6) \leq 872$). Houston's 872 and Egan–Houston's 5906 are ledgered as upper bounds only — **not** re-derived on box tonight.

Abstract

A *superpermutation* on n symbols is a string containing every permutation of those symbols as a contiguous substring; $s(n)$ is its minimal length (OEIS A180632). Classically $s(n) = \sum_{k=1}^{n} k!$ for $n \leq 5$. For $n \geq 6$ the factorial-sum conjecture fails (Houston 2014). This Band-1 probe:

1. Ledgers exact values and current bounds through $n = 7$.
2. Exhaustively searches minimal length for $n \leq 4$ (BFS, identity-fixed start).
3. Certificate-checks classical witnesses / recursive constructions through $n = 7$.
4. Marks $n = 6$ as the next open exact value.

Bound / value move this run: methods validation + literature frontier honesty. No new astronomical upper-bound record.

Problem statement

n	Standing $s(n)$ / bounds	Source class
1	**1**	trivial
2	**3**	brute force
3	**9**	brute force
4	**33**	brute force
5	**153**	Chaffin 2014 exhaustive
6	$868 \leq s(6) \leq 872$	Kropp LB; Houston UB — **OPEN exact**
7	$5884 \leq s(7) \leq 5906$	anon LB; Egan–Houston UB — **OPEN exact**

Classical constructive upper bound (always): $s(n) \le L(n) := \sum_{k=1}^{n} k!$.

Literature ledger (not all re-proved on box)

Item	Result	Citation / note
Ashlock–Tillotson construction	$s(n) \le \sum k!$	*Congressus Numerantium* 93 (1993)
Exact $n \le 4$	$1, 3, 9, 33$	classical brute force; Johnston overview
Exact $n = 5$	153; 8 distinct mins	Chaffin 2014
Conjecture fails $n \ge 6$	$s(6) \le 872 < 873$	Houston, arXiv:1408.5108
Anonymous lower bound	$s(n) \ge n! + (n-1)! + (n-2)! + n - 3$ $(n \ge 3)$	4chan / Houston–Pantone–Vatter (OEIS)
Improved $n = 6$ lower	$s(6) \ge 868$	Kropp (computer-assisted)
Williams–Egan L_2	$s(n) \le n! + (n-1)! + (n-2)! + (n-3)! + n - 3$ $(n > 3)$	Egan (adaptation of Williams 2013)
Record $n = 7$ upper	$s(7) \le 5906$	Egan–Houston 2019
Preliminary $a(6) = 872$ claims	**not** treated as archival-closed here	e.g. vlad-ds/a6-872 (conditional / preliminary)

Frontier conclusion (conservative): exact known for $1 \le n \le 5$; **next open exact is** $n = 6$.

Method (box)

1. **Certificate.** Given string S and n, extract every length-n window; check the set equals all $n!$ permutations.
2. **Recursive construction.** From a superpermutation on $n-1$ symbols, expand each first-appearance $(n-1)$-perm p to $p\,n\,p$ and squeeze maximal overlaps $\rightarrow$ length exactly $L(n)$.
3. **Exhaustive BFS ($n \le 4$).** State = (suffix of length $n-1$, bitmask of covered perms); fix start at identity $12\ldots n$ (relabeling symmetry). Extend by appending a symbol in $\{1, \ldots, n\}$. First time the mask is full $\Rightarrow$ minimal length. Separately BFS with limit $s - 1$ to confirm no shorter string under the same symmetry reduction.
4. $n = 5$**.** Certificate-check Johnston/Egan length-153 witness; minimality attributed to Chaffin (not re-exhausted; $5! = 120$ state space is heavy for naive BFS tonight).
5. $n \ge 6$**.** Construct and verify classical $L(n)$ upper witnesses only; ledger Houston / Egan–Houston records without claiming to reproduce TSP / kernel searches.

Results (numbers!)

Unit tests

`PASS`: known witnesses $n = 1..5$; recursive construction is a superpermutation of length $L(n)$ for $n = 1..7$; truncated $n = 3$ string correctly **rejected**.

Box BFS — exact minimal lengths

n	Box $s(n)$	Literature	No shorter (limit $s-1$)?	BFS nodes	Wall (s)
1	1	1	—	1	$< 10^{-4}$
2	3	3	yes	2 / 1	$< 10^{-4}$
3	9	9	yes	19 / 3	$< 10^{-4}$
4	33	33	yes	192188 / 41727	≈ 0.58 / 0.11

All four match literature. Witnesses recovered (identity-normalized) agree in length with the classical strings.

$n = 5$ **certificate**

- Witness length **153**; covers all **120** permutations: **yes**.
- Minimality: **literature** (Chaffin 2014) — not re-exhausted on box.

Recursive upper witnesses verified as superpermutations

n	Length $L(n)$	Is superpermutation?
1	1	yes
2	3	yes
3	9	yes
4	33	yes
5	153	yes
6	873	yes
7	5913	yes

Open / contested ledger (honesty)

- $n = 6$**:** $868 \le s(6) \le 872$. Box built 873 only. Houston's 872 **not** re-searched. Preliminary equality-to-872 packages **not** adopted as closed.
- $n = 7$**:** $5884 \le s(7) \le 5906$ (anon LB; record UB). Box built 5913 only. **No** claim of 5906.
- **Next open exact:** $n = 6$.

Total experiment wall ≈ 0.74 s.

Why Band-1

Finite exact values for tiny n are checkable; the asymptotic / large-n minimal-length landscape is open-ended research. Band-1 = ledger + small-n certificates. Claiming $s(6) = 872$ or a new $n = 7$ upper without a reproducible witness on box would violate the honesty stamp.

Reproducibility

- Code: `/workspace/experiments/band1/superpermutation/`
- Results: `results.json`

- Literature: `literature_ledger.json`
- Log: `/workspace/experiments/logs/2026-09-11_superpermutation.log`
- Run: `python3 superpermutation_search.py`

Conclusion

Band-1 deliverable: exact $s(1..4) = (1, 3, 9, 33)$ by box BFS; $s(5) = 153$ witness certified; classical $L(n)$ constructions verified through $n = 7$; literature frontier **open exact at** $n = 6$ with $868 \leq s(6) \leq 872$. **Does not** claim Houston 872 or Egan–Houston 5906 as box-produced records.